

Activity_Explore descriptive statistics

May 20, 2025

1 Activity: Explore descriptive statistics

1.1 Introduction

Data professionals often use descriptive statistics to understand the data they are working with and provide collaborators with a summary of the relative location of values in the data, as well as information about its spread.

For this activity, you are a member of an analytics team for the United States Environmental Protection Agency (EPA). You are assigned to analyze data on air quality with respect to carbon monoxide, a major air pollutant. The data includes information from more than 200 sites, identified by state, county, city, and local site names. You will use Python functions to gather statistics about air quality, then share insights with stakeholders.

1.2 Step 1: Imports

Import the relevant Python libraries `pandas` and `numpy`.

```
[9]: # Import relevant Python libraries.

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

The dataset provided is in the form of a `.csv` file named `c4_epa_air_quality.csv`. It contains a subset of data from the U.S. EPA. As shown in this cell, the dataset has been automatically loaded in for you. You do not need to download the `.csv` file, or provide more code, in order to access the dataset and proceed with this lab. Please continue with this activity by completing the following instructions.

```
[2]: # RUN THIS CELL TO IMPORT YOUR DATA.

### YOUR CODE HERE
epa_data = pd.read_csv("c4_epa_air_quality.csv", index_col = 0)
```

Hint 1

Refer to the video about loading data in Python.

Hint 2

There is a function in the `pandas` library that allows you to read in data from a `.csv` file and load it into a `DataFrame`.

Hint 3

Use the `read_csv` function from the `pandas` library. The `index_col` parameter can be set to 0 to read in the first column as an index (and to avoid "Unnamed: 0" appearing as a column in the resulting `DataFrame`).

1.3 Step 2: Data exploration

To understand how the dataset is structured, display the first 10 rows of the data.

```
[3]: # Display first 10 rows of the data.
```

```
epa_data.head(10)
```

```
[3]:
```

	date_local	state_name	county_name	city_name \
0	2018-01-01	Arizona	Maricopa	Buckeye
1	2018-01-01	Ohio	Belmont	Shadyside
2	2018-01-01	Wyoming	Teton	Not in a city
3	2018-01-01	Pennsylvania	Philadelphia	Philadelphia
4	2018-01-01	Iowa	Polk	Des Moines
5	2018-01-01	Hawaii	Honolulu	Not in a city
6	2018-01-01	Hawaii	Honolulu	Not in a city
7	2018-01-01	Pennsylvania	Erie	Erie
8	2018-01-01	Hawaii	Honolulu	Honolulu
9	2018-01-01	Colorado	Larimer	Fort Collins

	local_site_name	parameter_name \
0	BUCKEYE	Carbon monoxide
1	Shadyside	Carbon monoxide
2	Yellowstone National Park - Old Faithful Snow ...	Carbon monoxide
3	North East Waste (NEW)	Carbon monoxide
4	CARPENTER	Carbon monoxide
5	Kapolei	Carbon monoxide
6	Kapolei	Carbon monoxide
7	NaN	Carbon monoxide
8	Honolulu	Carbon monoxide
9	Fort Collins - CSU - S. Mason	Carbon monoxide

	units_of_measure	arithmetic_mean	aqi
0	Parts per million	0.473684	7
1	Parts per million	0.263158	5
2	Parts per million	0.111111	2
3	Parts per million	0.300000	3

4	Parts per million	0.215789	3
5	Parts per million	0.994737	14
6	Parts per million	0.200000	2
7	Parts per million	0.200000	2
8	Parts per million	0.400000	5
9	Parts per million	0.300000	6

Hint 1

Refer to the video about exploratory data analysis in Python.

Hint 2

There is a function in the **pandas** library that allows you to get a specific number of rows from the top of a DataFrame.

Hint 3

Use the **head()** function from the **pandas** library.

Question: What does the **aqi** column represent?

I believe that the “aqi” in this data set stands for the EPA’s “Air Quality Index(AQI)”.

Now, get a table that contains some descriptive statistics about the data.

```
[4]: # Get descriptive stats.

epa_data.describe()
```

```
[4]:
```

	arithmetic_mean	aqi
count	260.000000	260.000000
mean	0.403169	6.757692
std	0.317902	7.061707
min	0.000000	0.000000
25%	0.200000	2.000000
50%	0.276315	5.000000
75%	0.516009	9.000000
max	1.921053	50.000000

Hint 1

Refer to the video about descriptive statistics in Python.

Hint 2

There is a function in the **pandas** library that allows you to generate a table of basic descriptive statistics about the numeric columns in a DataFrame.

Hint 3

Use the **describe()** function from the **pandas** library.

Question: Based on the table of descriptive statistics, what do you notice about the count value for the **aqi** column?

The count value for the aqi column is 260. This means that there are 260 measurements represented in the data set.

Question: What do you notice about the 25th percentile for the aqi column?

This is an important measure for understanding where the aqi values lie.

The value of the 25th percentile column is 2. Meaning that 25% of the values in the dataset are below 2.

Question: What do you notice about the 75th percentile for the aqi column?

This is another important measure for understanding where the aqi values lie.

The value in the 75th percetile column is 9. Meanig that 75% of values in the dataset are above 9.

1.4 Step 3: Statistical tests

Next, get some descriptive statistics about the states in the data.

```
[7]: # Get descriptive stats about the states in the data.
```

```
epa_data['state_name'].describe()
```

```
[7]: count          260
     unique          52
     top      California
     freq           66
     Name: state_name, dtype: object
```

Hint 1

Refer to the video about descriptive statistics in Python.

Hint 2

There is a function in the **pandas** library that allows you to generate basic descriptive statistics about a DataFrame or a column you are interested in.

Hint 3

Use the **describe()** function from the **pandas** library. Note that this function can be used: - “on a DataFrame (to find descriptive statistics about the numeric columns)” - “directly on a column containing categorical data (to find pertinent descriptive statistics)”

Question: What do you notice while reviewing the descriptive statistics about the states in the data?

Note: Sometimes you have to individually calculate statistics. To review to that approach, use the **numpy** library to calculate each of the main statistics in the preceding table for the aqi column.

There are 260 state values, and 52 of them are unique. The most commonly occurring state name in the dataset is Calafornia, with a frequency of 66. Meaning that 66 values in the dataset correspond to aqi measurements taken in California.

1.5 Step 4. Results and evaluation

Now, compute the mean value from the `aqi` column.

```
[10]: # Compute the mean value from the aqi column.  
  
np.mean(epa_data['aqi'])
```

```
[10]: 6.757692307692308
```

Hint 1

Refer to the video about descriptive statistics in Python.

Hint 2

There is a function in the `numpy` library that allows you to get the mean value from an array or a Series of values.

Hint 3

Use the `mean()` function from the `numpy` library.

Question: What do you notice about the mean value from the `aqi` column?

This is an important measure, as it tells you what the average air quality is based on the data.

The mean value for the `aqi` column is approximately 6.76 (rounded to 2 decimal places). This means that the average `aqi` from the data is approximately 6.76.

Next, compute the median value from the `aqi` column.

```
[11]: # Compute the median value from the aqi column.  
  
np.median(epa_data['aqi'])
```

```
[11]: 5.0
```

Hint 1

Refer to the video about descriptive statistics in Python.

Hint 2

There is a function in the `numpy` library that allows you to get the median value from an array or a series of values.

Hint 3

Use the `median()` function from the `numpy` library.

Question: What do you notice about the median value from the `aqi` column?

This is an important measure for understanding the central location of the data.

The median value in the `aqi` column is 5.0. This means that half of the `aqi` values are below 5.

Next, identify the minimum value from the `aqi` column.

```
[12]: # Identify the minimum value from the aqi column.  
  
np.min(epa_data['aqi'])
```

```
[12]: 0
```

Hint 1

Refer to the video about descriptive statistics in Python.

Hint 2

There is a function in the `numpy` library that allows you to get the minimum value from an array or a Series of values.

Hint 3

Use the `min()` function from the `numpy` library.

Question: What do you notice about the minimum value from the `aqi` column?

This is an important measure, as it tell you the best air quality observed in the data.

The minimum value in the `aqi` column is 0. Meaning that 0 is the smallest `aqi` value in the dataset.

Now, identify the maximum value from the `aqi` column.

```
[13]: # Identify the maximum value from the aqi column.  
  
np.max(epa_data['aqi'])
```

```
[13]: 50
```

Hint 1

Refer to the video about descriptive statistics in Python.

Hint 2

There is a function in the `numpy` library that allows you to get the maximum value from an array or a Series of values.

Hint 3

Use the `max()` function from the `numpy` library.

Question: What do you notice about the maximum value from the `aqi` column?

This is an important measure, as it tells you which value in the data corresponds to the worst air quality observed in the data.

The maximum value in the `aqi` column is 50. Meaning that 50 is the biggest `aqi` value in the dataset.

Now, compute the standard deviation for the `aqi` column.

By default, the `numpy` library uses 0 as the Delta Degrees of Freedom, while `pandas` library uses 1. To get the same value for standard deviation using either library, specify the `ddof` parameter to 1 when calculating standard deviation.

```
[15]: # Compute the standard deviation for the aqi column.  
  
np.std(epa_data['aqi'], ddof=1)
```

```
[15]: 7.0617066788207215
```

Hint 1

Refer to the video section about descriptive statistics in Python.

Hint 2

There is a function in the `numpy` library that allows you to get the standard deviation from an array or a series of values.

Hint 3

Use the `std()` function from the `numpy` library. Make sure to specify the `ddof` parameter as 1. To read more about this function, refer to its documentation in the references section of this lab.

Question: What do you notice about the standard deviation for the `aqi` column?

This is an important measure of how spread out the `aqi` values are.

The standard deviation for the `aqi` column is 7.06 (rounded to 2 decimal places). This is the measure of how spread out the `aqi` values are in the dataset.

1.6 Considerations

What are some key takeaways that you learned during this lab?

Functions in the `pandas` and `numpy` libraries can be used to find statistics that describe a dataset. The `describe()` function from `pandas` generates a table of descriptive statistics about numerical or categorical columns. The `mean()`, `median()`, `min()`, `max()` and `std()` functions from `numpy` are useful for finding individual statistics about numerical data.

How would you present your findings from this lab to others? Consider the following relevant points noted by AirNow.gov as you respond: - “AQI values at or below 100 are generally thought of as satisfactory. When AQI values are above 100, air quality is considered to be unhealthy—at first for certain sensitive groups of people, then for everyone as AQI values increase.” - “An AQI of 100 for carbon monoxide corresponds to a level of 9.4 parts per million.”

The average AQI value in the data is approximately 6.67, which is considered safe with respect to carbon monoxide. Further, 75% of the AQI values are below 9.

What summary would you provide to stakeholders? Use the same information provided previously from AirNow.gov as you respond.

- 75% of the AQI values in the data are below 9, which is considered good air quality.

- Funding should be allocated for further investigation of the less healthy regions in order to learn how to improve the conditions.

References

[Air Quality Index - A Guide to Air Quality and Your Health](#). (2014,February)

[Numpy.Std — NumPy v1.23 Manual](#)

US EPA, OAR. (2014, 8 July).*Air Data: Air Quality Data Collected at Outdoor Monitors Across the US*.

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